

Global CO₂ Emissions: Who Emits, How Much, and What's Changing

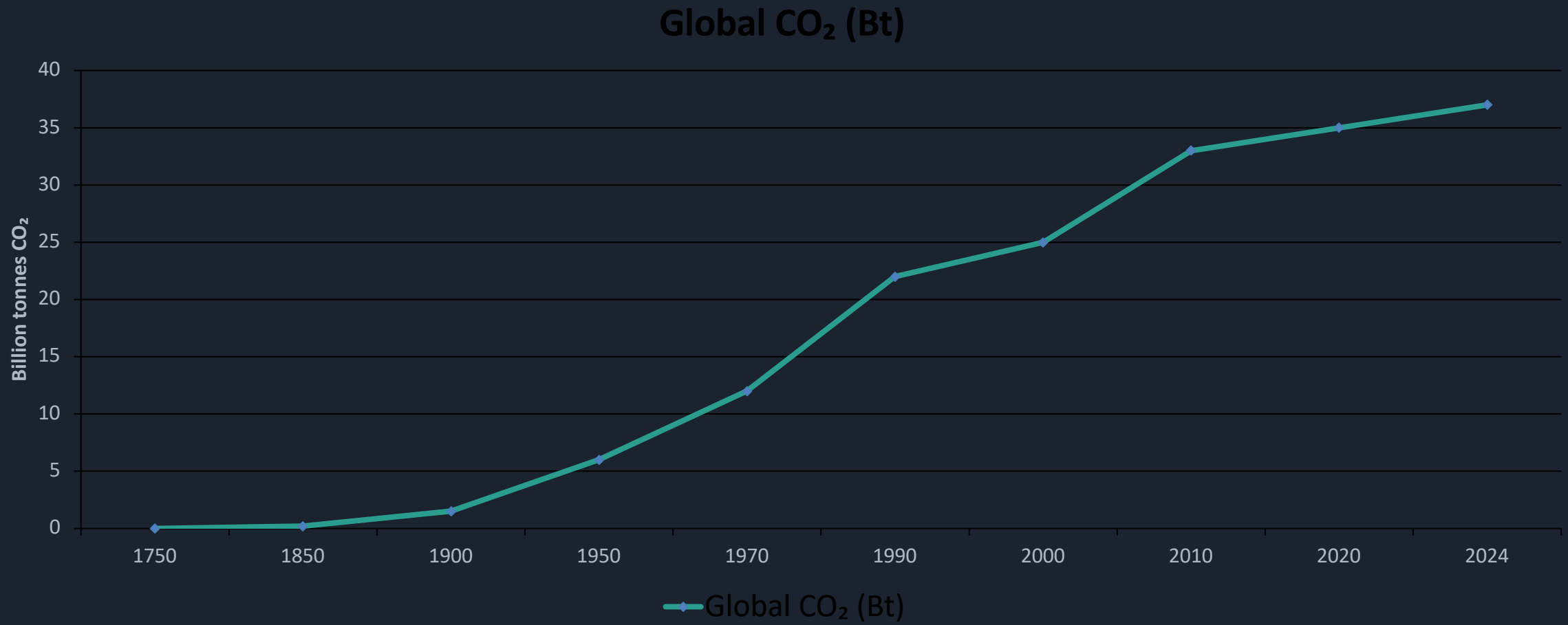
Fossil fuels & industry · 1750 – 2024

For policy analysts, climate negotiators, and sustainability researchers

Data source: Global Carbon Budget (2025) via Our World in Data

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

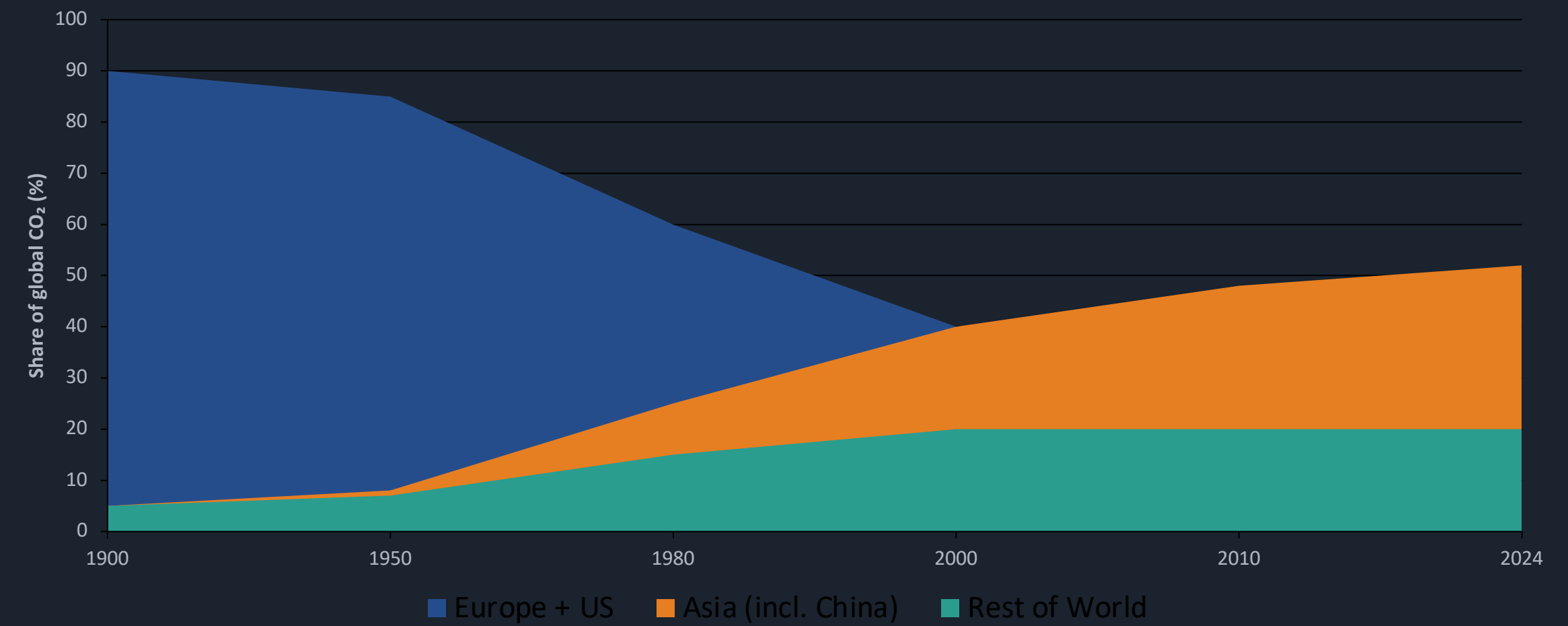
Long-run trajectory of fossil fuel & industry CO₂, 1750–2024



Key inflections: post-1950 acceleration; growth slowing since ~2010, but no peak yet.

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

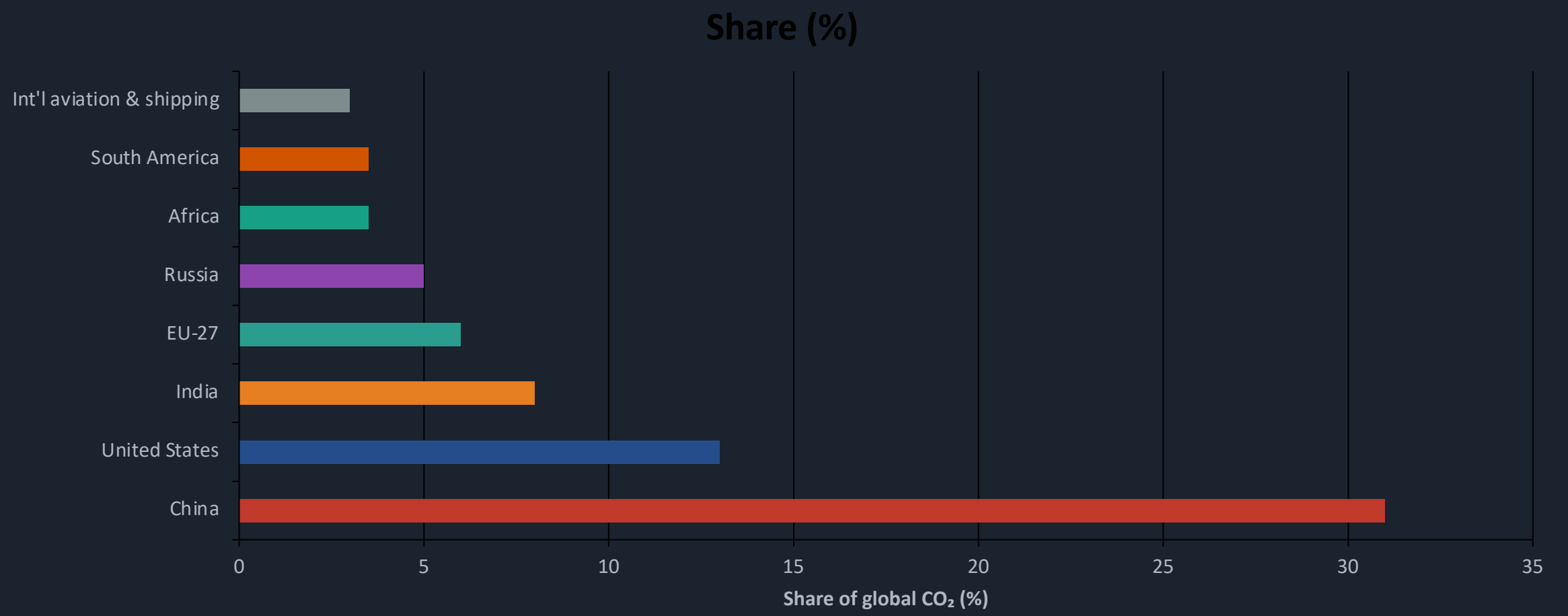
Rise of Asia (especially China) as Europe and North America's share declined



In 1900, Europe and the US accounted for >90%; today Asia produces roughly half of global emissions.

China Now Emits More Than One-Quarter of Global CO₂

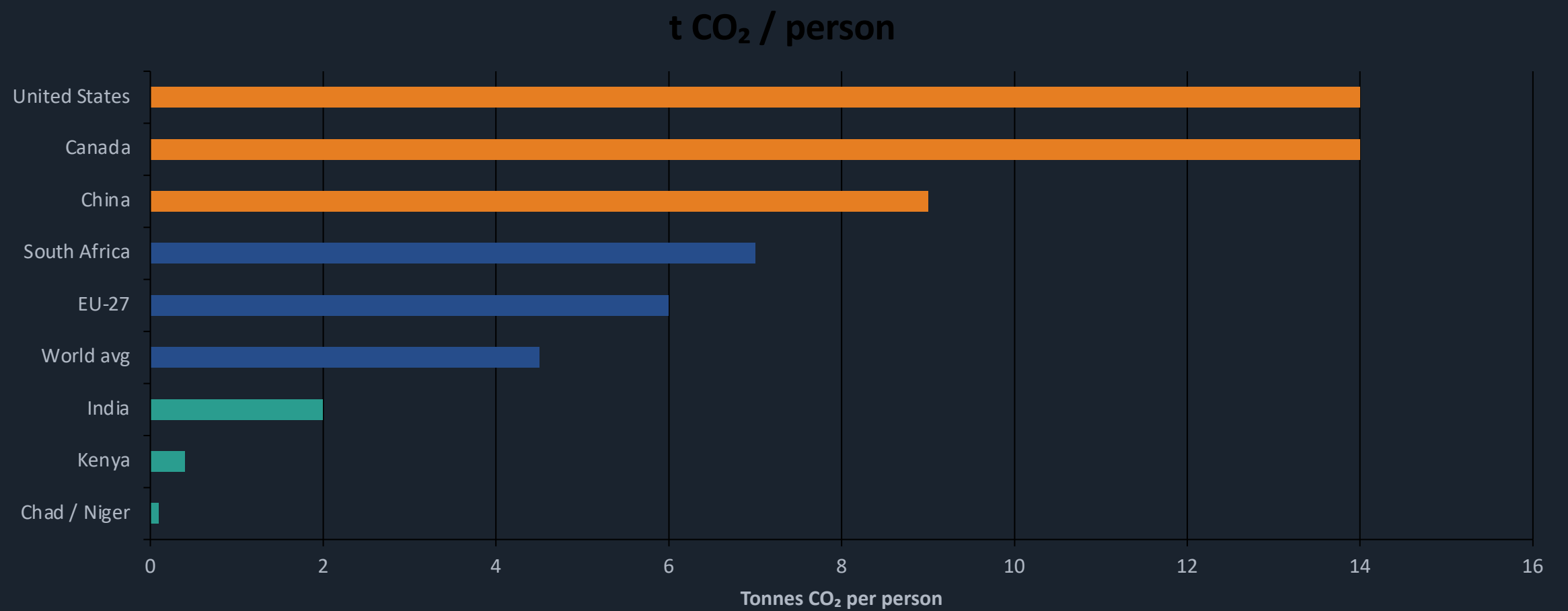
Country-level shares of global CO₂ emissions, 2024



China ~31% | US ~13% | India ~8% | EU-27 ~6% | Russia ~5% | Africa & South America each ~3–4%

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

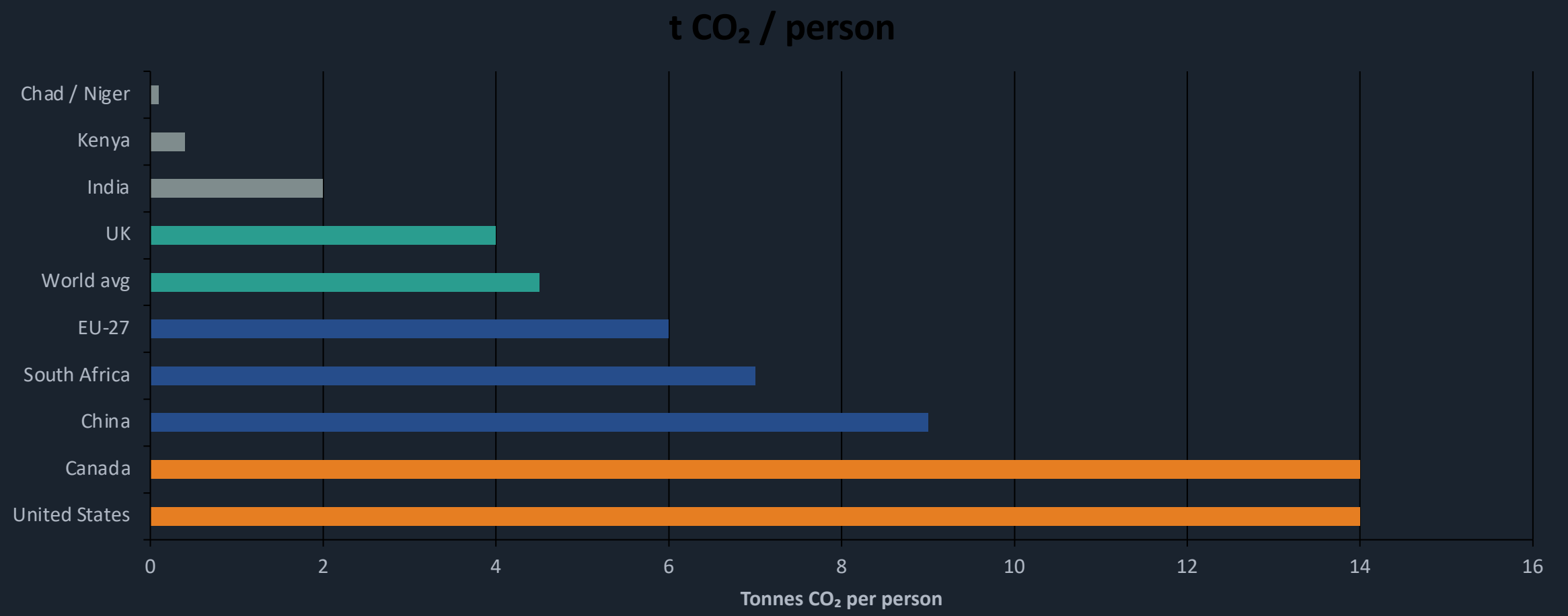
A 150x gap between the lowest and highest large-country emitters



An American or Australian emits in under two days what a person in Mali or Niger emits in a year.

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

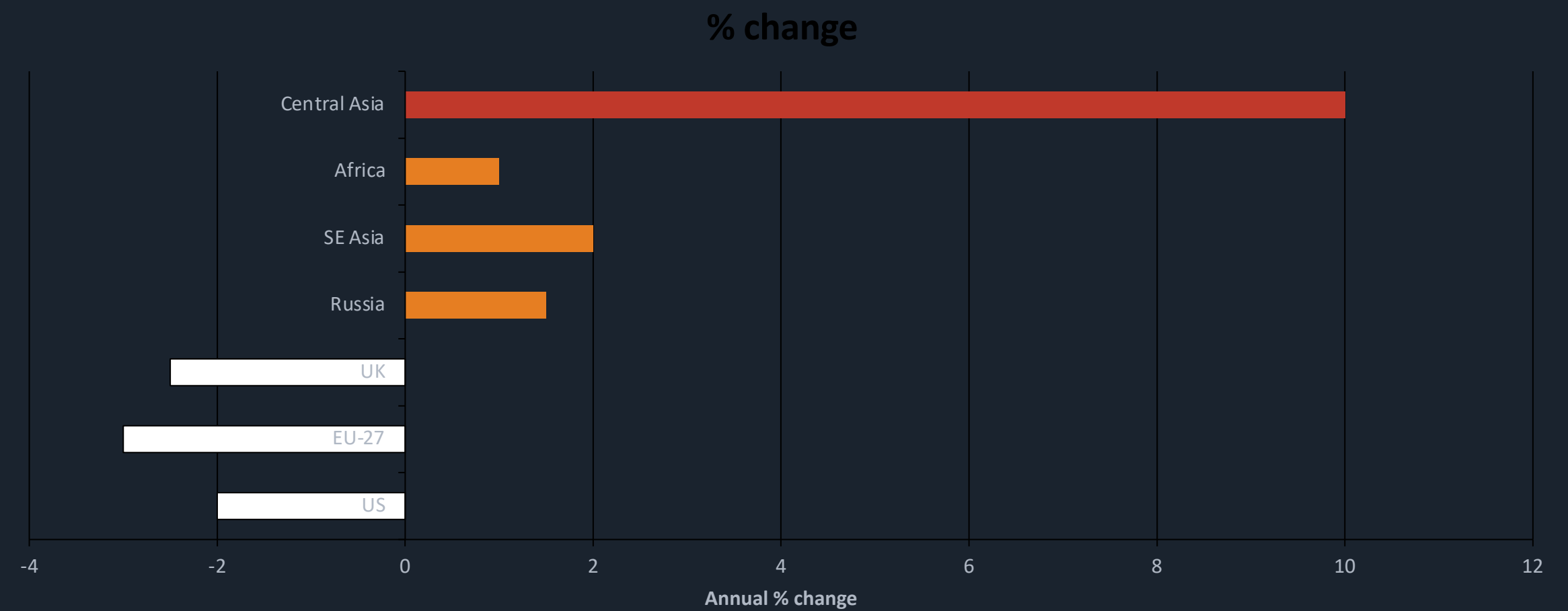
Current per capita CO₂ emissions for key countries and regions, 2024



China's per capita emissions remain below the US despite being the largest total emitter; India is very low at ~2 t.

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

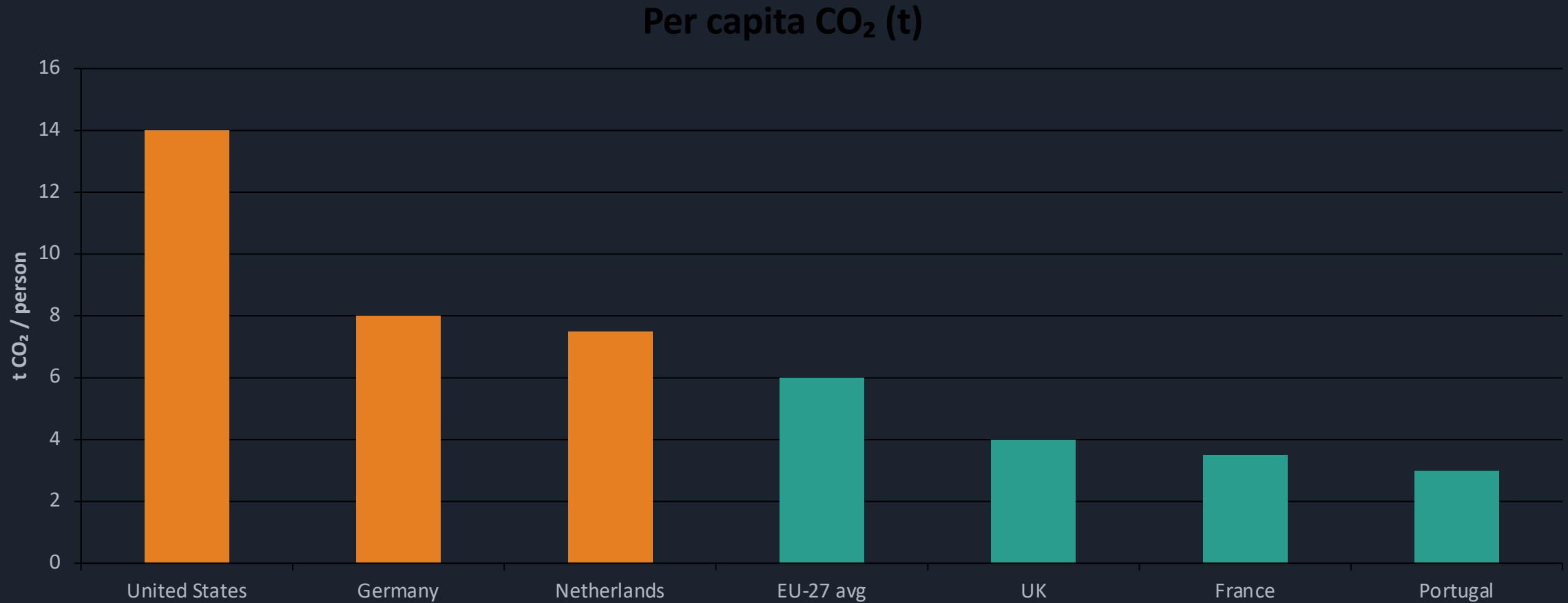
Annual percentage change in CO₂ emissions, 2024



Blue = decline (US, EU, UK); Orange/Red = increase (Russia, SE Asia, Africa, Central Asia >10%).

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Among wealthy nations, electricity sources drive large differences in per capita emissions



- France, the UK, and Portugal rely more on nuclear and renewables, cutting per capita emissions sharply.
- Germany, the Netherlands, and the US remain more fossil-dependent, keeping emissions higher.
- Similar prosperity levels — very different carbon footprints, driven by policy and technology choices.

Key Takeaways: Responsibility Is Shared but Unequal

What the data tell us about climate action and equity

1. No peak yet — Global fossil CO₂ emissions exceed 35 Bt/yr and have not peaked, despite slowing growth.
2. China is the largest annual emitter, but its per capita emissions (~9 t) remain well below the US (~14 t).
3. Historical responsibility is dominated by Europe and the US (>90% in 1900), even as their shares have fallen.
4. Per capita gaps of ~150× persist between the richest nations (US/Canada ~14 t) and the poorest (Chad/Niger ~0.1 t).
5. Energy policy matters — countries with similar prosperity can have very different footprints depending on technology choices.